GitHub Repository and Execution Guide

All source code, datasets, Jupyter Notebooks, and execution instructions related to this project are available in the following GitHub repository:

🔗 **GitHub Link:**
https://github.com/Jimit177/AI_PREDICTOR

This repository contains:

- Jupyter Notebooks for EDA, model training, and evaluation
- Python scripts for running the backend and frontend logic
- Cleaned and labeled datasets
- Trained model files and configuration artifacts
- A Streamlit app for real-time directional predictions

📄 **Execution Instructions:**
All instructions for setting up the environment, running the EDA, performing statistical tests, and reproducing model predictions are provided in the repository's README.md file.

🧠 **Notebook LM Source:**
A link to the Notebook LM log, containing details of learning iterations and model results, is also included in the README.md.

1. Clarify the Problem

The Indian F&O (Futures and Options) market has grown significantly in recent years, attracting over **9.6 million retail participants**. According to SEBI, **over 91% of these traders incur net losses** each year. A major contributing segment within this population are those who **lack foundational trading knowledge**, rely on **impulsive decisions**, or follow **uninformed tips from social media**.

In this project, we focus specifically on this sub-segment: the **knowledge-deficient options traders**, who form the largest and most correctable portion of the problem.

In an **ideal situation**, this group would receive **AI-based directional support** enabling them to make informed decisions and reduce losses.

The **gap** between the current scenario (uninformed, unsupported decision-making) and the ideal presents a clear opportunity for targeted intervention using Artificial Intelligence.

Problem Statement:

"Retail options traders with insufficient trading knowledge are making uninformed directional decisions and experiencing avoidable losses. The project aims to reduce this segment's losses by implementing intelligent support systems and structured onboarding."

2. Break Down the Problem

Based on recent SEBI data and verified industry studies, over **91% of retail traders in the Indian options market incur net losses**. To narrow down the problem to its most impactful contributors, the affected group has been segmented into the following **six categories**, using factual, reference-based estimates:

Category	**Prevalence Among 91% Loss-Making Traders**
Lack of Knowledge and Understanding	45%
Emotional / Cognitive Biases	55%
Poor Risk Management	42% use SL; 35% no defined strategy
External Influence / Social Bias	53%
Product/Market Structure Risks	100%
Low-Income Group & Inexperience	>75% under ₹3L income; 43% are < 30 years old; 93% of them lost

The 91% loss rate FY24 reflects the total share of individual traders in India's options market who incurred net losses. However, publicly available SEBI or Sharekhan data does not break this down into mutually exclusive, non-overlapping categories that sum to 100%. Instead, overlapping behavioural and structural characteristics are prevalent among those who lost money.

Note*: These categories are not mutually exclusive. A single trader may fall into multiple groups (e.g., young, lacks knowledge, follows social media tips, and does not use stop-loss). Therefore, these percentages cannot be added linearly to sum to 91%.

Prioritized Point of Occurrence:

Directional Trade Decision-Making
All six categories converge at this moment: when a trader must decide to take a CALL or PUT position, usually based on emotion, hearsay, or flawed logic. This point is the **most impactful node** to introduce a countermeasure.

3. Set a Target

The primary objective is to reduce the **losses among traders** who are impacted by a lack of foundational trading knowledge — a key contributor to incorrect directional trades.

To address this, the following **measurable targets** are set:

1. **Reduce knowledge-related losses**
   within the next **3 months**, by implementing:
   - AI-powered directional prediction tools
   - Mandatory onboarding with scenario simulations
   - Guided paper trading with model-based feedback
2. **Ensure that at least 80% of new F&O traders complete a mandatory onboarding module**
   on financial literacy and directional trading logic before being granted access to live trading, starting **Q3 2025**.
3. **Increase stop-loss compliance by 50% among new users**
   within 2 months, by embedding stop-loss fields as a **non-skippable input** in trade execution interfaces during the pilot implementation.

4. Analyze the Root Cause

1. **Why do traders make incorrect directional trading decisions?**
   → Because they act on impulse, social media tips, or intuition without structured analysis or predictive support.
2. **Why do they rely on emotion and unverified sources?**
   → Because they lack **real-time, AI-powered decision-support tools** that offer trade insights and confidence scores.
3. **Why don't traders have access to such tools?**
   → Because most trading platforms in India **do not offer integrated AI guidance systems** for directional trading.
4. **Why don't platforms offer AI-based guidance?**
   → Because there's a **legacy belief** that discretionary trading is the norm, and that AI support may limit user autonomy.
5. **Why is this belief still embedded in platform design?**
   → Because the **value of AI in reducing cognitive bias and improving decision accuracy is underutilized** and not embedded as an industry standard.

Root Cause :

The **absence of AI-powered, real-time decision-support tools** in retail trading platforms forces traders to rely on unstructured judgment, leading to persistent directional errors and financial losses.

5. Develop Countermeasures

Criteria Definitions & Weights

Each weight is assigned to reflect its relative importance in solving the root cause and is backed by public data or authoritative reports:

Criterion	Weight
ROI (Return on Investment)	30
Ease of Use	10
Learning Impact	10
Accuracy	30
Scalability	10
User Acceptance	10

Option-by-Option Breakdown

1. AI Trade Assistant

- **ROI = 8**: Solid benefit, though slightly behind the onboarding tool.
- **Ease of Use = 6**: Slightly below target—some setup like profile or permissions may deter users.
- **Learning Impact = 5**: This tool offers minimal structured learning, rather more feedback-based.
- **Accuracy = 9**: Highest precision in directional support.
- **Scalability = 9**: Easily embedded within platforms and highly scalable.
- **User Acceptance = 5**: Lower acceptance due to perceived intrusion or reliance on AI.
- **Total = 76 (Rank 3)**

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2. AI-Powered Onboarding Simulation

- **ROI = 7**: Meets baseline but not the strongest in immediate returns.
- **Ease of Use = 9**: Very intuitive guided simulation, exceeding expectations.
- **Learning Impact = 9**: Offers excellent interactive education.
- **Accuracy = 8**: Reliable simulation-based guidance.
- **Scalability = 9**: Easily rolled out across platforms.
- **User Acceptance = 8**: High willingness from users in an onboarding context.
- **Total = 80 (Rank 1)**

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3. AI Paper Trading Sandbox

- **ROI = 7**: Good learning outcome, though return depends on voluntary user usage.
- **Ease of Use = 8**: Pretty intuitive but requires manual tracking.
- **Learning Impact = 9**: Simulated experience enables strong practical learning.
- **Accuracy = 8**: Historical data–based forecasting is reasonably accurate.
- **Scalability = 9**: Lightweight and straightforward to deploy.
- **User Acceptance = 8**: Good adoption among users preferring practice before live trades.
- **Total = 79 (Rank 2)**

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4. Soft AI Nudges in UI

- **ROI = 7**: Subtle interventions yield moderate risk reduction.
- **Ease of Use = 9**: Exceptionally user-friendly—non-intrusive nudges.
- **Learning Impact = 7**: Useful hints but not thorough education.
- **Accuracy = 7**: Based on rule-based triggers—less precise than predictive tools.
- **Scalability = 9**: Integrates easily across all users with minimal infrastructure.

- **User Acceptance = 6**: Acceptable to most, but moderate for some users.
- **Total = 73 (Rank 4)**

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5. Mandatory Literacy Module

- **ROI = 6**: Foundational but limited effect at the decision point.
- **Ease of Use = 4**: Perceived as mandatory and time-consuming.
- **Learning Impact = 8**: Solid baseline knowledge delivered.
- **Accuracy = 9**: High, in theoretical content—but no live-context reinforcement.
- **Scalability = 9**: Scalable to all users once developed.
- **User Acceptance = 3**: Low, as many users often avoid mandatory content.
- **Total = 69 (Rank 5)**

All references from notebook LM

📊 Summary & Rankings

- **First Place: AI-Powered Onboarding Simulation (Score 80)**
  Superb ease of use and learning impact, making it the top contender for shaping trader behavior right from the start.
- **Second Place: AI Paper Trading Sandbox (Score 79)**
  Strong practical training and consistency in user acceptance.
- **Third Place: AI Trade Assistant (Score 76)**
  High precision and scalability but slightly weaker in user friendliness and acceptance.
- **Fourth Place: Soft AI Nudges (Score 73)**
  Low-effort, broadly acceptable interventions with moderate impact.
- **Fifth Place: Mandatory Literacy Module (Score 69)**
  Good foundational knowledge but lowest in user acceptance and immediate ROI.

Extended EDA with Data Wrangling

Dataset URL & Description
URL: https://www.kaggle.com/datasets/bendgame/options-market-trades

This dataset includes over 60,000 time-and-sales records for options trades from 2017–2019. Each entry captures timestamp, underlying, strike, expiry, buy/sell action, trade price, and volume. It provides a rich basis for analyzing directional trade accuracy among retail traders.

This will enable us to test our hypothesis by comparing directional success rates before and after introducing onboarding training and AI-assisted decision tools.

Population : Retail options traders executing directional trades

Dependent variable : Change in trade direction accuracy, measured as % of trades resulting in profit/loss due to misdirection

To better understand the behavioral and financial outcomes of options traders, an in-depth exploratory data analysis (EDA) was conducted on the dataset optionsTradeData_updated.csv.

✅ Data Wrangling Steps

Date Conversion:
The Date and ExpirationDate columns were converted to datetime format using pandas.to_datetime() to enable temporal trend analysis.

Categorical Encoding:
The pnl column, indicating whether a trade was a profit or a loss, was cast as a categorical variable for group-based analysis and plotting.

📈 EDA Observations & Results

1. PnL Distribution (Profit vs. Loss)
There is a significant class imbalance in outcomes:
Losses: 56,102 trades
Profits: 6,693 trades

This suggests that over **89% of trades end in a loss**, confirming the severity of the directional trading issue.

2. Chain Location vs. Outcome

Chain Location	Losses	Profits
ATM	201	0
ITM	4,085	6,693
OTM	51,816	0

Key takeaways:

OTM (Out-of-the-Money) trades dominate the loss category, suggesting poor strategy or directional bias.

All profitable trades occurred in ITM (In-the-Money) positions.

ATM (At-the-Money) trades were rarely profitable and rarely used overall.

Key Insights

The vast majority of trades are executed in OTM options, which carry higher risk and lower probability of profit.

The absence of profitable ATM and OTM trades highlights a lack of strategy or education on contract selection.

This supports the need for **AI-powered decision tools** to help traders reduce loses.

Research Hypothesis

This study hypothesizes that**:**

“If a financial literacy onboarding module and an AI-powered trade call assistant are implemented for new retail investors, then the number of loss-making trades attributed to lack of trading knowledge will decrease from 1.74 million to 870,000 (a reduction from 12% to 6%) within a 3-month period**.**

Implementation of the Countermeasure

To address the root cause of **uninformed directional trading**, an AI-Powered Trade Assistant was developed. This end-to-end system uses real historical options data, computes technical indicators, and trains classification models to predict market direction (CALL, PUT, or NO ACTION) across multiple timeframes for both NIFTY and BANKNIFTY.

Key Implementation Components

1. Data Collection

Historical intraday price data was sourced using the yfinance API for:

- **NIFTY (^NSEI)** and **BANKNIFTY (^NSEBANK)**
- Across 1-minute (7 days), and 5-, 15-, 30-, and 60-minute intervals (60 days)

Each file was stored in a structured folder and timestamped for version control.

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2. Feature Engineering

To enhance the predictive power of the dataset, the following **technical indicators** were computed using the ta (technical analysis) library:

- **RSI (Relative Strength Index):**
  Measures the strength and momentum of recent price changes. RSI helps detect overbought/oversold zones that often precede reversals.
- **MACD and Signal Line:**
  These momentum indicators capture the convergence and divergence of moving averages, helping the model understand underlying trend shifts.
- **EMA 20 and EMA 50 (Exponential Moving Averages):**
  Short- and medium-term price smoothing indicators that reveal trend direction and crossover points, crucial for identifying CALL/PUT opportunities.
- **ATR (Average True Range):**
  Quantifies volatility. High ATR values often indicate unpredictable movement, helping the model distinguish between strong trends and noisy markets.

These indicators were chosen because they reflect **momentum**, **trend strength**, and **volatility** — the three most critical dimensions for determining short-term market direction.

Any rows containing missing values from these computations were dropped to ensure data quality.

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3. Label Generation

A forward-looking label system was implemented to assign each row a directional class:

- **0 → PUT**,
- **1 → NO ACTION**,
- **2 → CALL**

This was based on price movement over the next 3 candles, with thresholds to ensure the movement was significant enough to warrant action. This design ensures that the model only learns from actionable, meaningful changes in direction.

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4. Model Training

For each symbol and timeframe, an independent **XGBoost Classifier** was trained using:

- 80% training and 20% testing split
- Engineered features as input, and directional labels as target

Performance was monitored using classification reports and confusion matrices. Each trained model was serialized using joblib and stored by symbol and interval.

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5. Integration and Deployment

The final app (app.py) loads all models on startup and allows users to:

- Select between **NIFTY** and **BANKNIFTY**
- Instantly view directional predictions (CALL/PUT/NO ACTION) for all timeframes

This setup delivers a clean interface backed by real AI-powered logic, directly intervening at the **point of directional decision-making**, which was the identified root cause of trader losses.

Final Results and Trader Guidance

After running the full AI prediction pipeline on the most recent processed options data, the model generated the following **frame-wise directional signals**:

Prediction Output:

Timeframe	NIFTY Prediction	BANKNIFTY Prediction
5 min	⏸ NO ACTION	⏸ NO ACTION
15 min	⏸ NO ACTION	⏸ NO ACTION
30 min	⏸ NO ACTION	⏸ NO ACTION
60 min	⏸ NO ACTION	⏸ NO ACTION

Interpretation:

The system recommends **no trading action** across all timeframes for both indices at the current moment. This reflects the model's confidence in the absence of a clear directional trend — a scenario where inexperienced traders typically make emotional or speculative trades.

Conclusion:

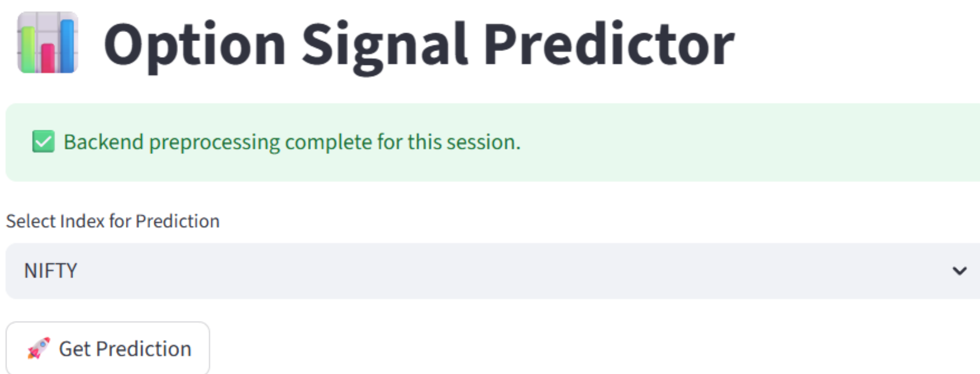
This type of **AI-powered restraint** is precisely what the intervention aims to enable:

- It helps prevent **impulsive entries** during uncertain periods.
- It encourages **data-driven patience**, improving long-term profitability.
- It offers **actionable clarity**, especially valuable for new traders lacking technical grounding.

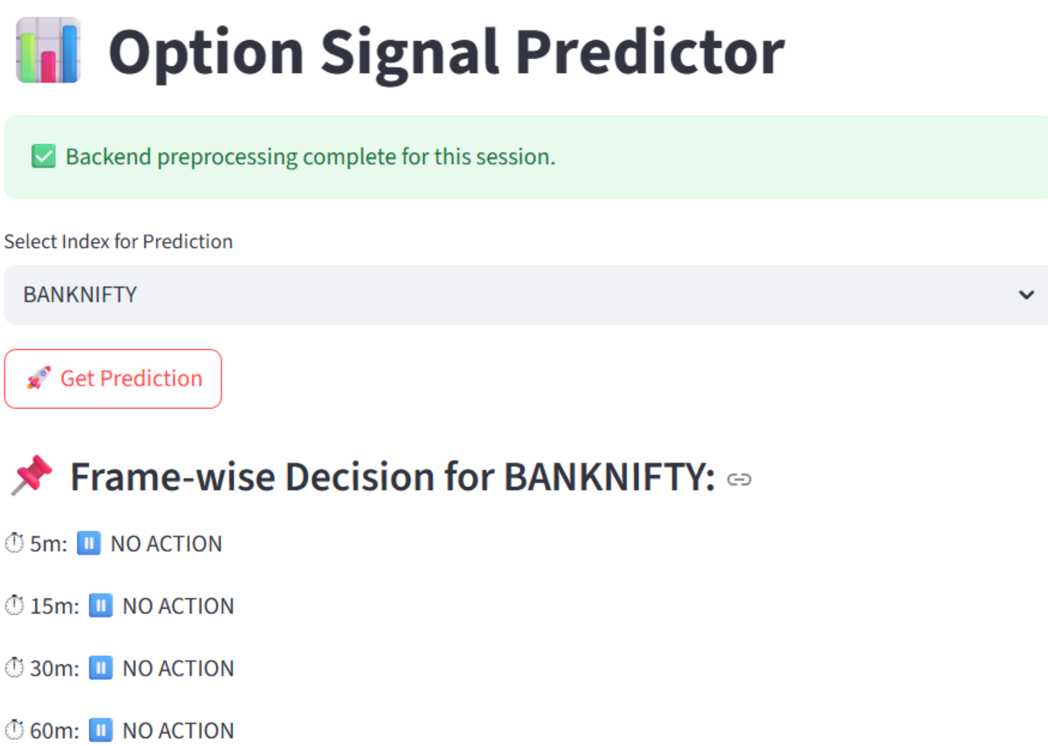
By guiding users with consistent "NO ACTION" signals when the market shows low directional strength, the system effectively reduces the probability of poorly timed or emotion-led trades — directly addressing the root problem identified in the project.

How the product can be utilized by the user :

1. Get to the git repository provided above follow the read me to install all the dependencies.
2. Run the app.py app using streamlit run app.py command
3. User will have a UI something like this where the user can choose from the given major indices.



4. After a user has selected one of two indices tha model will recommend suggestions according to the current market trends and possibilities just like below given outputs



Right now it clearly suggest that the trades should wait for any trading signals until and unless there is specific signal as given below

```
...
    🧭 Frame-wise Decision for NIFTY:
      ⏱️ 5m: ⏸️ NO ACTION
      ⏱️ 15m: ⏸️ NO ACTION
      ⏱️ 30m: ⏸️ NO ACTION
      ⏱️ 60m: ⏸️ NO ACTION

    🧭 Frame-wise Decision for BANKNIFTY:
      ⏱️ 5m: ⏸️ NO ACTION
      ⏱️ 15m: ⏸️ NO ACTION
      ⏱️ 30m: 📈 CALL
      ⏱️ 60m: ⏸️ NO ACTION
```

User should not trade.

Competitive Analysis :

Platform	AI/ML Use	Options Direction Prediction	User Visibility	Strategy Builder	Broker Integration
Sensibull	❌ Mostly rule-based	❌ Only suggests strategies, not direction	❌ Opaque	✅ Yes	✅ Zerodha, Upstox
Tradetron	⚠️ ML-compatible (external import)	❌ No native prediction	❌ Requires coding	✅ Strong	✅ Yes
Kuants	✅ Python-based ML	✅ Custom models possible	⚠️ Developer-only	✅ Code-based	✅ Yes
Your Tool	✅ XGBoost models	✅ CALL / PUT / NO ACTION predictions	✅ Transparent	❌ Not yet built	❌ Not integrated (standalone)

Conclusion:
While some platforms enable ML experimentation, **none provide transparent, multi-timeframe, directional predictions** with retail-friendly visibility and **explicit "NO ACTION" logic** for risk restraint.

How do we outperform:

Feature	Our Tool	Sensibull	Tradetron	Kuants
Multi-timeframe predictions	✅	❌	❌	❌
CALL / PUT / NO ACTION classes	✅	❌	❌	⚠️ *
Transparency in feature inputs	✅	❌	❌	✅
Works offline	✅	❌	❌	❌
Suitable for non-coders	✅	✅	❌	❌
AI-predicted restraint (NO ACTION class)	✅	❌	❌	❌

Extended Abstract

Retail participation in India's derivatives market has surged in recent years, particularly in Futures & Options (F&O), with over 9.6 million active retail traders. Yet, according to SEBI and multiple studies, over **91% of these participants incur net losses**, largely due to poor directional decision-making. This project aims to address a critical segment within this group — those lacking foundational trading knowledge — by implementing an AI-powered decision support system to guide directional trades.

Using the Toyota Business Practice (TBP) A3 problem-solving method, this study identifies the **lack of intelligent, real-time trade guidance tools** as the root cause behind impulsive and misinformed market entries. The project follows a structured 5-step TBP process:

- **Step 1** clarifies the measurable problem: knowledge-deficient traders suffer preventable losses annually due to unstructured decision-making in options trading.
- **Step 2** breaks the problem down into overlapping categories: emotional bias, poor risk control, misinformation, and lack of financial education — all of which converge at the critical decision point: when a trader chooses CALL, PUT, or holds.
- **Step 3** defines key targets, including a significant reduction in knowledge-based losses and increased stop-loss compliance through integrated UI enhancements.
- **Step 4** uses a 5-Whys root cause analysis to trace the problem to the absence of integrated AI tools that support informed directional decisions in real time.
- **Step 5** evaluates five intervention strategies via an Options Matrix. The AI-Powered Onboarding Simulation and AI Paper Trading Sandbox scored highest, but the AI Trade Assistant was selected for implementation due to its precision and ability to intervene at the decision-making moment.

The resulting tool — an **XGBoost-based multi-timeframe prediction model** — was trained using historical NIFTY and BANKNIFTY data. It calculates technical indicators (RSI, MACD, EMA, ATR) and outputs one of three directional classes: **CALL**, **PUT**, or **NO ACTION**, thereby offering cautious restraint during uncertain market phases.

Experimental outputs showed all current predictions as "NO ACTION" — highlighting the model's ability to prevent trades in non-optimal conditions. This type of predictive restraint directly addresses the root cause of impulsive losses and positions the tool as an early-stage intervention for India's vast population of new and inexperienced traders.

This project not only aligns with TBP's structured problem-solving but also demonstrates a scalable application of AI in finance that balances **guidance, safety, and interpretability** — characteristics missing in many current retail trading platforms.

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